

Clinical-sonographic scores for the screening of placenta accreta spectrum: a systematic review and meta-analysis



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OBJECTIVE: Clinical-sonographic scoring systems combining clinical features and ultrasound imaging markers have been proposed for the screening of placenta accreta spectrum, but their usefulness in different settings remains limited. This study aimed to assess and compare different clinical-sonographic score systems applied from mid-pregnancy for the prenatal evaluation of patients at risk of placenta accreta spectrum at birth.

DATA SOURCES: PubMed/MEDLINE, Google Scholar, and Embase were searched between October 1982 and October 2022 to identify eligible studies.

STUDY ELIGIBILITY CRITERIA: Observational studies providing data on the use of a combined clinical-ultrasound score system applied from mid-pregnancy for the prenatal evaluation of placenta accreta spectrum were included.

METHODS: Study characteristics were evaluated by 2 independent reviewers using a predesigned protocol registered on PROSPERO (CRD42022332486). Heterogeneity among studies was analyzed with Cochran's Q-test and I^2 statistics. Statistical heterogeneity was quantified by estimating the variance between the studies using I^2 statistics. The area under the receiver operating characteristic curve of each score and their summary receiver operating characteristic curves were calculated with sensitivity and specificity, and the integrated score of the summaries of the receiver operating characteristic curves of all sonographic markers was calculated. Forest plots were used to develop the meta-analysis of each sonographic marker and for the integrated sonographic score.

RESULTS: Of 1028 articles reviewed, 12 cohorts and 2 case–control studies including 1630 patients screened for placenta accreta spectrum by clinical-ultrasound scores met the eligibility criteria. A diagnosis of placenta accreta spectrum was reported in 602 (36.9%) cases, for which 547 (90.9%) intraoperative findings and/or histopathologic data were described. A wide variation was observed among the studies in reported sensitivities and specificities and in thresholds used for the identification of patients with a high probability of placenta accreta spectrum at birth. The summaries of the areas under the curve of the individual sonographic scores ranged from 0.85 (the lowest) for subplacental hypervascularity to 0.91 for placental location in the lower uterine segment, myometrial thinning, and placental lacunae and 0.95 for the loss of clear zone. Only 4 studies included placental bulging in their sonographic score system, and therefore no meta-analysis for this score was performed. The integrated summary of the areas under the curve was 0.83 (95% confidence interval, 79–0.86). Forest plot analysis revealed integrated sensitivities and specificities of 0.68 (95% confidence interval, 0.53–0.80) and 0.88 (95% confidence interval, 0.68–0.96), respectively.

CONCLUSION: Clinical-sonographic score systems can contribute to the prenatal screening of patients at risk of placenta accreta spectrum at birth. Although we included multiple sonographic studies conducted during the mid-pregnancy period, standardized evaluation should be

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performed not only with strict ultrasound criteria for the placental position, mid third trimester gestational age at examination, and sonographic markers associated with PAS. Numeric sensitivities, specificities, NPVs, PPV, LR-, and LR+ should be recorded prospectively to assess their accuracy in different set-ups and PTP should be verified at delivery. The variables recommended for most predictive screening are: loss of clear zone underneath the placental bed, placentation in the LUS, and placenta lacunae.

Key words: cesarean delivery, meta-analysis, placenta accreta spectrum, scores, systematic review, ultrasound imaging

Introduction

Placenta accreta spectrum (PAS) was first reported in 1982.¹ However, epidemiologic studies have found that until recently, PAS remained undiagnosed before birth in between half^{2,3} and two-thirds of cases.⁴ The role of ultrasound imaging in the prenatal screening of patients at risk of PAS at birth is well established⁵ and highlighted in the recommendations of all current guidelines.^{6–10}

Cesarean delivery (CD) is associated with an increased incidence of placentation in the lower uterine segment (LUS), and is the main clinical marker associated with the development of accreta placentation in subsequent pregnancies.^{10,11} The individual and combined risks for CD and accreta placentation increase with the number of previous CDs, and pregnant patients

with a history of multiple CDs, presenting with a low-lying placenta or placenta previa, have the highest risk of PAS at birth.¹² When unsuspected cases are identified at the time of delivery, attempts to manually remove accreta placental tissue can be associated with major and uncontrollable intraoperative bleeding.^{6,7} It is now recommended that pregnant patients with these characteristics have a detailed ultrasound examination of their placenta for changes associated with accreta placentation.^{6–10,13}

Although ultrasound imaging has a high accuracy rate in identifying patients with a high probability of PAS at birth when performed by experts, the corresponding expertise is not widely available, and thus missed diagnosis is common.^{13,14} In 2016, we proposed a probability scoring system combining

the number of previous CDs and sonographic markers associated with PAS,¹⁵ which we tested using targeted ultrasound protocols for pregnant patients at risk of PAS. We found that the implementation of the scoring system improved the detection rate of patients with PAS at birth.¹⁶

Objective

This study aimed to assess and compare different clinical-sonographic score systems in the prenatal evaluation of patients at risk of PAS at birth.

Methods

Search strategy

We searched PubMed/MEDLINE, Google Scholar, and Embase for studies published in English. We started with the study of Tabsh et al,¹ which was published in October 1982, and ended with studies published by October 2022 (40 years). The *a priori* designed protocol complies with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines,¹⁷ and was registered on PROSPERO (CRD42022332486) (www.crd.york.ac.uk/PROSPERO). The overall search strategy included free-text headings and MeSH (Medical Subject Headings) terms for free-text words and word variants of “placenta accreta,” “placenta increta,” “placenta percreta,” “abnormally invasive placenta,” “morbidity adherent placenta,” and “placenta previa accreta,” combined with “ultrasound imaging” OR “ultrasound scores” OR “ultrasound index” OR “ultrasound screening” OR “prenatal diagnosis.”

Selection criteria

Studies had to meet the following criteria to be included in the review: (1)

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Why was this study conducted?

Patients with a history of cesarean delivery presenting with a placenta located in the lower uterine segment (LUS) are at high risk of placenta accreta spectrum (PAS) and should be screened for ultrasound markers associated with PAS at birth. This study aimed to investigate the contribution of combined clinical-sonographic score systems in the prenatal evaluation of PAS.

Key findings

Clinical-sonographic score systems that include the placental location in the LUS, absence of clear zone, and presence of placental lacunae have the highest sensitivity and specificity. The variables that are the most recommended for predictive screening are loss of clear zone underneath the placental bed, placentation in the LUS, and placental lacunae.

What does this add to what is known?

Clinical-sonographic score systems may contribute to the screening and triage of patients at high risk of PAS at birth. However, these scores should be further evaluated prospectively using a standardized protocol that includes the placental location and gestational age at the mid-point of the third-trimester examination.

observational study (cohort or case–control) that investigates clinical-sonographic score systems combining a clinical history of ≥ 1 previous CDs and sonographic markers used in the prenatal assessment of PAS; and (2) study reporting on the outcome of patients with singleton pregnancies with PAS confirmed at birth. We excluded case reports, case series of <10 patients, letters, editorials, reviews, non-English-language articles, and duplications of previously published data from the same centers. We also excluded multiple pregnancies because the diagnosis of placenta accreta in twins and triplet pregnancies is characterized by many unique challenges, which are beyond the scope of our study and might complicate the interpretation of the results.

Previous CD is the most common etiology for subsequent PAS. Because of their rarity, cases of PAS due to causes other than previous CD were omitted from this study, and should be addressed in additional studies.

We are aware that magnetic resonance imaging (MRI) has a potential contribution to the diagnosis of PAS. However, given that not all hospitals have this technology, we prefer to increase the awareness of the clinicians using the most common sonographic markers for the screening of PAS and for generating the risk score for PAS.

Recently, there has been increased interest in screening for PAS during the first half of the pregnancy. Further large prospective studies are needed to validate the extrapolation of these sonographic markers for earlier screening of PAS in advanced gestational age (GA), and therefore these studies were omitted from the current analysis.

Screening strategy

Two reviewers (M.P.Z. and P.E.R.) independently analyzed the search results. Selected studies were evaluated in 2 stages. First, the abstracts of all potentially relevant articles were individually examined for eligibility. These were then obtained in full text and independently assessed for content, data extraction, and analysis. Additional relevant studies were identified from

reference lists of reviews and editorials. A third investigator (E.J.) resolved disagreements between the 2 original reviewers.

Data extraction

A predesigned data extraction form was used by 2 reviewers (M.P.Z. and P.E.R.) and included authors' names, country, year of publication, GA at sonographic assessment, recruitment setting, sample size, demographics (parity, gravidity, maternal age, and body mass index [BMI]), and pregnancy history (previous CDs, placenta previa, etc.).

The ultrasound markers of patients with previous CD who are at risk of PAS at delivery have been already visualized and described in recent studies and guidelines, including a recent state-of-the-art review.^{5,8,18} Our meta-analysis reviewed the scoring systems of the following sonographic markers: (1) placental position in the LUS,^{5,8,18} (2) loss of the hypoechoic retroplacental zone underneath the placental bed (loss of clear zone),^{5,8} (3) "ballooning" of the uterus containing the placenta into the surrounding pelvic structure and bladder wall interruption (placental bulging),^{5,8} (4) myometrial thickness <1 mm or undetectable myometrial thickness (myometrial thinning),^{5,8} (5) presence of large, hypervascular uteroplacental and bridging vessels with irregular hypoechoic (without a hyperechoic halo) intraplacental spaces located above large feeder vessels, giving the placenta a "moth-eaten" appearance (lacunae),¹⁸ and (6) considerable amount of color Doppler signals seen in the placental bed, demonstrating multidirectional flow and aliasing artefact (subplacental hypervascularity).^{5,8,18} All studies included in our analysis had at least 3 of the above 6 markers.

The histopathologic findings at birth were recorded. All search results were combined in a reference database. Duplicates were removed. Disagreements between the 2 original reviewers were resolved by discussion with the third investigator (R.M.). Individual sonographic markers with their individual sensitivities and specificities, and all other cohort characteristics were

compared and then integrated into the meta-analysis.

Quality and risk of bias assessment of the studies

Risk of bias was assessed in 4 main domains using QUADAS-2, a revised tool for assessing patient selection, index test, reference standard, flow, and timing.¹⁹ The quality indicators assessed were the study design and the conduct and analysis of all included studies. Two independent reviewers (M.P.Z. and P.E.R.) scored each item as "high," "low," or "unclear" if there was insufficient information to make an accurate judgment on the risk for bias. No study was excluded on the basis of the risk of bias assessment. A third reviewer (E.J.) resolved discrepancies of the evaluation. We assessed the following criteria: main study characteristics, the type of the used ultrasound markers associated with a high probability of PAS at birth, and the number of patients included in each score for accurately quantifying the sensitivity and specificity of PAS diagnosis at birth.

Data synthesis

For the meta-analysis, 2 reviewers not previously involved in creating the individual studies' database (A.S.N. and H. M.) made independent assessments of the included studies to reduce the risk of bias for each study. The sensitivity and specificity of each parameter in each study were calculated on the basis of the number of PAS cases at birth with true positive (TP), true negative (TN), false positive (FP), and false negative (FN) for each study.

Positive and negative likelihood ratios (LR+, LR−) and posttest probability (PTP) were calculated according to Akobeng (2007) in the following manner²⁰:

$$LR+ = \text{sensitivity} / (1 - \text{specificity})$$

$$LR- = (1 - \text{sensitivity}) / \text{specificity}$$

PTP calculation:

$$1. \text{Prevalence} = (TP + FN) / (TP + FN + FP + TN)$$

$$2. \text{Pretest odds} = \text{prevalence} / (1 - \text{prevalence})$$

$$3. \text{Posttest odds} = \text{pretest odds} \times LR+$$

4. PTP=posttest odds/(1+posttest odds)

Forest plots. Forest plots were used to visualize the pooled estimates for each ultrasound marker according to their sensitivities and specificities. We quantified the statistical heterogeneity by estimating the variance between the studies using I^2 statistics, where the values of I^2 range between 0 and 100%. I^2 between 25% and <50% was considered to indicate low heterogeneity, I^2 between 50% and <75% moderate heterogeneity, and I^2 >75% substantial heterogeneity. Given that heterogeneity of the values used to prepare the forest plot is expected a priori, we estimated the pooled sensitivity and specificity effect and their 95% confidence interval (CI) using the random effects of the model. The heterogeneity between studies was calculated by chi-square test. The study's relative size is indicated by the symbol size of each plot, and then for the integrated sonographic scores of all studies.

Area under the receiver operating characteristic curve. Receiver operating characteristic (ROC) curves were plotted and areas under the curve (AUCs) were calculated from the sensitivity and specificity of each sonographic marker. The summaries of AUCs (SAUC) and ROC curves (SROC) for each sonographic marker were the means of the markers' AUC and ROC across all studies, with each study weighted according to its proportion of the total patient population across all studies reporting that marker.

To create the integrated scoring system, the mean SAUC of SROCs of the individual sonographic markers was calculated. The SAUC of SROC of each marker was weighted according to the proportion of studies that measured this sonographic marker compared with the total number of studies for all markers.

Stata, version 15 (StataCorp LLC, College Station, TX) was used to analyze all the data and plots. A P value <.05 was considered significant.

Results

Study selection and characteristics

A total of 1028 articles met the criteria of the search terms used in the literature

search. Of the 1028 studies, there were 12 cohort studies and 2 case-control studies (Table 1^{21–34}) that included at least 3 of the 6 sonographic markers of PAS outlined in the methods section. The 14 studies that met the eligibility criteria included 1630 patients who were screened for PAS (process of selection summarized in Supplemental Figure S1). Six studies reported morbidity associated with the detection of PAS,^{24,26,28,30,33} and 6 studies reported on focal PAS.^{22,28,30–33,35}

Risk of bias

Results of the QUADAS-2 assessment are provided in Supplemental Figure S2. In terms of risk of bias and concerns regarding applicability, all selected studies in the patient domain included patients with a history of ≥ 1 previous CDs and placentation in the LUS. Five studies were considered to have high risk of bias because they did not describe the use of transvaginal scans (TVS) to confirm the placental position in the LUS.^{23,25,28,33–34} Another 5 studies were considered to have an unclear risk because they did not include patients presenting with loss of clear zone.^{22,25,32–34} In the index-test domain, 3 studies were considered at high risk because they did not refer to standardized ultrasound scores associated with PAS at birth,^{21,26,28} and 4 were unclear because they did not describe the position of the placenta (ie, anterior or posterior).^{24,29–31} In terms of the reference-standard domain, most studies defined the presence of abnormal attachment of placental tissue to the uterine wall, and basic histologic descriptions served as a clear reference standard. Five of these studies also reported on the grade of PAS (ie, accreta, increta, or percreta).^{23,24,29,32,33} One study was listed as high-risk because the authors did not provide data on both intraoperative gross findings and microscopic examination.³¹ In addition, 2 studies were also considered to have an unclear risk of bias because they did not report the histologic examination.^{28,34} The analysis included 3 studies with 85 PAS cases for which no histological verifications were

available.^{19,20,23} Concerning the flow-and-timing domain, the time elapsed between the index test and the reference standard was >4 weeks in studies that reported data on GA at ultrasound examination and at birth.^{23,27,29,32} Three studies were considered at high risk of bias because the reported GA at the analysis covered the entire second and third trimester of pregnancy, and the authors did not report the GA at birth.^{24,28,31}

Study characteristics

The characteristics of the corresponding studies are presented in Table 2.^{21–34} The listed studies were conducted in 8 different countries between 2008 and 2022.^{21–34} Except 3 studies,^{22,29,32} all included publications were prospective. Two publications had a case-control design,^{21,29} and the others were cohort studies. Nine studies were from a single center,^{22–25,28–29,31–33} and the remaining 5 were from tertiary referral centers. A diagnosis of PAS was reported in 602 (36.9%) of the evaluated patients. The size of the studies ranged between 46 and 210 patients, and the incidence of PAS cases at birth ranged between 13.6% to 63%, confirming that these studies included patients with a high risk of PAS at birth (Table 2^{21–34}). The SAUC of the published studies, as reported by their authors, was >0.84 for all studies and increased >0.90 for studies with 5 to 6 ultrasound scores (Table 2^{21–34}).

Table 2^{21–34} displays the sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), LR+, LR–, and PTP of each sonographic marker in each individual study. Three studies included all of the 6 sonographic markers.^{21,30–31} Four studies had only 3 sonographic markers.^{22,27,33,34} Four studies included 4 markers,^{23–24,32} and 2 included 5 markers.^{28,30} Accordingly, the score of placental bulging was measured in only 4 studies,^{21,28,29,31} and thus, an individual forest plot or SAUC of SROC was not calculated for this sonographic marker in the meta-analysis (Table 3). A large variability was also found in the

TABLE 1
Characteristics of the included studies

Study	Year	Country	Study period	Type of study and design	PAS cases/cohort (n)	Description of LUS placentation/GA	TAS and TVS	Histopathology IOP macro/micro
Wong et al ²¹	2008	New Zealand	Jan. 2004–Dec. 2006	Prospective case–control	9/66 (13.6%)	All cases placenta previa/second and third trimesters	Yes	No/Yes
Rac et al ²²	2015	USA	Dec. 1997–Dec. 2011	Retrospective cohort	54/184 (29.4%)	All cases low-lying or placenta previa/third trimester	Yes	No/Yes
Maged et al ²³	2018	Egypt	Jan. 2015–Jan. 2016	Prospective cohort	63/100 (63%)	All cases anterior placenta previa total or partial ^a /28 wk	No TVS	Yes/Yes
Cali et al ²⁴	2018	Italy	2007–2017	Prospective cohort	57/210 (27.1%)	All cases anterior placenta previa/second and third trimesters	Yes	Yes/Yes
Chong et al ²⁵	2018	China	Jan. 2015–Oct. 2017	Prospective cohort	112/137 (81.8%)	All cases of placenta previa/second and third trimesters	No TVS	No/Yes
Pătru et al ²⁶	2019	Romania	Jan. 2016–Dec. 2017	Prospective cohort	11/46 (23/9%)	All cases anterior placenta previa covering the IO (41 cases in third trimester)	Yes	Yes/Yes
Zhu and Xie ²⁷	2019	China	Dec. 2016–Mar. 2018	Prospective cohort	36/51 (70.6%)	All cases anterior placenta covering previous CD scar ^a /≥28 wk	Yes	Yes/Yes
Barati et al ²⁸	2020	Iran	2015–2017	Prospective cohort	39/150 (26%)	All cases low-lying or placenta previa ^a /second and third trimesters	No TVS	Yes/No
Del Negro et al ²⁹	2020	Italy	Jan. 2014–Nov. 2019	Retrospective case–control	19/138 (13.8%)	All cases low-lying or placenta previa/third trimester	Yes	Yes/Yes
El-Haieg et al ³⁰	2021	Egypt	July 2016–Mar. 2018	Prospective cohort	38/114 (33.3%)	All cases anterior placenta previa/≥28 wk	Yes	Yes/Yes
Gulati et al ³¹	2021	India	Nov. 2015–Mar. 2017	Prospective cohort	19/69 (27.5%)	All cases low-lying or placenta previa/≥24 wk	Yes	N/A
Happe et al ³²	2021	USA	Jan. 2012–Dec. 2018	Retrospective cohort	92/194 (47.4%)	All cases anterior low-lying or placenta previa/≥28 wk	Yes	Yes/Yes
Abu Hashim et al ³³	2022	Egypt	July 2017–Dec. 2018	Prospective cohort	23/100 (23%)	All cases anterior low-lying or placenta previa/third trimester	No TVS	Yes/Yes
Bansal et al ³⁴	2022	India	Oct. 2018–Mar. 2020	Prospective cohort	30/71 (42.3%)	All cases anterior placenta previa/third trimester	No TVS	Yes/No

CD, cesarean delivery; GA, gestational age; IO, internal os; IOP, intraoperative; LUS, lower uterine segment; N/A, not available; PAS, placenta accreta spectrum; TAS, transabdominal scan; TVS, transvaginal scan.

^a No standardized description of the ultrasound criteria used for placental location.

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PTP for each sonographic marker, as shown in [Table 2](#).^{21–34}

Synthesis of the summaries of the receiver operating characteristic curves and forest plots for the meta-analysis

To conduct the meta-analysis, the individual numeric values of 13 studies for the sensitivity, specificity, PPV, NPV, LR+, LR–, and PTP were extracted. These were not available for the Chong et al²⁵ study, which was thus not included in the meta-analysis ([Table 2](#)^{21–34}).

[Table 3](#) shows the sensitivity, specificity, SAUC, and the sensitivities at 90% and 80% specificities for 5 sonographic markers (excluding the placental bulging that was measured in only 4 studies). These were derived from the individual forest plots made for each sonographic marker ([Supplemental Figures S3–S7](#)) except placental bulging (because it was measured in only 4 publications).

The integrated forest plot of the scoring system ([Figure 1](#)) included all 6 sonographic markers, and had a sensitivity of 0.68 (95% CI, 0.53–0.80), a heterogeneity index (I^2) of 97% ($P<.001$), and a specificity of 0.88 (95% CI, 0.68–0.96), with a heterogeneity index $I^2=93%$ ($P<.001$). The P values for all sensitivities and specificities were statistically significant ($P<.001$). An integrated scoring system made of all 6 individual sonographic markers (including placental bulging) is presented at the bottom of [Table 3](#) and in [Figure 1](#). The lowest SAUC was found for subplacental hypervascularity (0.66; $n=9$) and myometrial thinning (0.78; $n=7$). When these 2 sonographic markers were removed from the forest plot, we gained only a 4% increase in both sensitivity (0.72 [0.52–0.86]) and specificity (0.92 [0.63–0.99]) ([Table 3](#)). The corresponding heterogeneity index I^2 for the sensitivity was increased by 0.15%, and the heterogeneity index I^2 for specificity by 1.68% ([Supplemental Figure S8](#)).

The comparison between the integrated sensitivities of 6 vs 4 sonographic markers at 90% specificity

TABLE 2

Description of the statistical performance of the ultrasound markers of the included studies

Study, year		Placentation in the LUS	Loss of clear zone	Placental bulge	Myometrial thinning	Placental lacunae	Subplacental hypervascularity	Number of ultrasound scores used	SAUC of the authors (95% CI)
Wong et al, ²¹ 2008	Sens	89.0	90.0	0.0	22.2	77.8	62.5		
	Spes	96.5	35.1	100.0	100.0	38.6	49.0	6	0.94
	PPV	80.0	19.6	—	100.0	16.7	16.1		(0.81–1.06)
	NPV	70.4	100.0	86.4	89.1	91.7	89.3		
	LR+	25.4	1.39	—	—	1.27	1.23		
Rac et al, ²² 2015	LR–	0.11	0	1.00	0.78	0.56	0.76		
	PTP	80.0	18.0			16.7	14.5		
	Sens	70.4				51.9	64.8		0.87
	Spec	72.3				87.7	81.5	3	
	PPV	51.4				63.6	59.3		(0.80–0.85)
	NPV	85.5				81.4	84.8		
	LR+	2.54				4.22	3.50		
	LR–	0.41				0.55	0.43		
Maged et al, ²³ 2018	PTP	51.3				63.7	59.2		
	Sens	82.5	87.3			93.7	47.6		
	Spes	81.1	89.1			62.2	94.6	4	N/A
	PPV	88.1	93.2			80.8	93.8		
	NPV	73.2	80.5			85.2	51.5		
	LR+	4.37	8.01			2.48	8.81		
	LR–	0.22	0.14			0.10	0.55		
Cali et al, ²⁴ 2018	PTP	88.2	93.2			80.9	93.7		
	Sens	81	90.5		100.0	79.0		1.00	
	Spes	100	81.0		81.9	100.0	4	(0.93–1.00)	
	PPV	100	82.6		84.7	100.0			
	NPV	84	98.5		100.0	82.7			
	LR+	—	4.76		5.52	—			
	LR–	0.19	0.12		0	0.21			
Chong et al, ²⁵ 2018 ^a	PTP	—	82.6		84.7	—			
	Sens							0.94	
	Spes							(N/A)	
	PPV								
	NPV								
	LR+								
	LR–								
Pătru et al, ²⁶ 2019	PTP								
	Sen	66.7	87.5	90.9	90.0				
	Spes	57.1	66.7	66.7	60.0		4	N/A	
	PPV	66.7	77.8	90.9	81.8				
	NPV	57.1	80.0	66.7	75.0				
	LR+	1.55	2.63	2.73	2.25				
	LR–	0.58	0.19	0.14	0.17				
	PTP	73.6	77.8	90.9	84.9				

(continued)

TABLE 2

Description of the statistical performance of the ultrasound markers of the included studies (continued)

Study, year		Placentation in the LUS	Loss of clear zone	Placental bulge	Myometrial thinning	Placental lacunae	Subplacental hypervascularity	Number of ultrasound scores used	SAUC of the authors (95% CI)
Zhu and Xie, ²⁷ 2019	Sens	63.9	27.8		80.6			0.89	
	Spes	33.3	100.0		46.7		3	(0.79–0.98)	
	PPV	69.7	100.0		78.4				
	NPV	27.8	36.6		50.0				
	LR+	0.96	—		1.51				
	LR–	1.08	0.72		0.42				
	PTP	69.7	—		78.4				
Barati et al, ²⁸ 2020	Sens	100.0	87.2	59.0		92.0	10.0		
	Spes	91.0	80.0	100.0		20.0	90.0	5	N/A
	PPV	80.0	94.0	100.0		82.0	80.0		
	NPV	100.0	62.0	38.0		40.0	20.0		
	LR+	11.11	4.35	—		1.15	1.00		
	LR–	0	0.16	0.41		0.40	1.00		
	PTP	97.7	94.4	—		81.8	79.6		
Del Negro et al, ²⁹ 2020	Sens	13.8	82.8	10.3	58.6	100.0	14.0		0.94
	Spes	100.0	100.0	100.0	97.2	100.0	100.0	6	(N/A)
	PPV	100.0	100.0	100.0	85.0	100.0	100.0		
	NPV	81.3	95.6	80.7	90.0	100.0	81.3		
	LR+	—	—	—	20.93	—	—		
	LR–	0.86	0.17	0.90	0.43	0	0.86		
	PTP	—	—	—	84.8	—	—		
El-Haieg et al, ³⁰ 2021	Sens	26.0	95.0		95.0	74.0	79.0		0.96,
	Spes	100.0	88.0		91.0	87.0	66.0	5	(0.91–1.00)
	PPV	100.0	80.0		84.0	74.0	54.0		
	NPV	73.0	97.0		97.0	87.0	86.0		
	LR+	—	7.92		10.56	5.69	2.32		
	LR–	0.74	0.06		0.05	0.30	0.32		
	PTP	—	79.8		84.1	74.0	53.7		
Gulati et al, ³¹ 2021	Sens	89.5	94.7	52.6	94.7	89.5	84.2		0.93
	Spes	83.9	93.5	100.0	96.8	64.5	93.5	6	(0.79–0.99)
	PPV	77.3	89.9	100.0	94.8	60.7	88.8		
	NPV	92.9	96.6	77.7	96.7	90.9	90.6		
	LR+	5.56	14.57	—	29.59	2.52	12.95		
	LR–	0.13	0.06	0.47	0.05	0.16	0.17		
	PTP	77.3	89.9	—	94.8	60.7	88.8		
Happe et al, ³² 2021	Sens	85.7	12.0	76.1	89.1		0.94		
	Spes	34.3	23.5	79.4	24.5	4	(0.91–0.97)		
	PPV	54.1	12.4	76.9	51.5				
	NPV	72.9	22.8	78.6	71.4				
	LR+	1.30	0.16	3.69	1.18				
	LR–	0.42	3.74	0.30	0.44				
	PTP	54.0	12.6	76.9	51.6				

(continued)

TABLE 2

Description of the statistical performance of the ultrasound markers of the included studies (continued)

Study, year		Placentation in the LUS	Loss of clear zone	Placental bulge	Myometrial thinning	Placental lacunae	Subplacental hypervascularity	Number of ultrasound scores used	SAUC of the authors (95% CI)
Abu Hashim et al, ³³ 2022	Sens	100.0		91.3	95.6		0.84		
	Spec	93.9		54.5	70.1	3	(0.75–0.91)		
	PPV	82.1		37.5	48.9				
	NPV	100.0		95.4	98.2				
	LR+	16.39		2.01	3.20				
	LR–	0		0.16	0.06				
	PTP	83.0		37.5	48.9				
Bansal et al, ³⁴ 2022	Sens	90.0	93.3	73.3			0.88		
	Spes	53.7	82.9	100.0		3	(N/A)		
	PPV	58.7	80.0	100.0					
	NPV	88.0	94.4	83.7					
	LR+	1.94	5.46	—					
	LR–	0.19	0.08	0.27					
	PTP	58.7	80.0	—					

The studies are listed with Sens, Spes, PPV, NPV, LR+, LR–, and PTP for each sonographic marker in each study. The integrated SAUC is the value that the authors published for the integrated scores with their 95% CIs.

CI, confidence interval; LR+, positive likelihood ratio; LR–, negative likelihood ratio; LUS, lower uterine segment; N/A, not available; NPV, negative predictive value; PPV, positive predictive value; PTP, posttest probability; SAUC, summary of the area under the receiver operating characteristic curve; Sens, sensitivity; Spes, specificity.

^a Values we calculated the statistical values from raw data presented in the cited paper, which the authors did not present there.

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showed that the sensitivity increased from 66% to 75%. At 80% specificity, the sensitivity increased from 75% to 81% (Table 3).

The SROC values of each individual sonographic marker of PAS (Table 3; Figure 2) were between 0.85 and 0.92, indicating a large diversity. The combined SAUC of all studies in the scoring system was 0.83 (95% CI, 0.79–0.86). The lowest SAUC was found for subplacental hypervascularity (0.85; 95% CI, 0.81–0.88; 10 studies). Myometrial thinning had an SAUC of 0.91 (95% CI, 0.88–0.93; 7 studies). The SAUC for placentation in the LUS was 0.91 (95% CI, 0.89–0.93; 13 studies), the SAUC for loss of clear zone was 0.95 (95% CI, 0.71–0.99; 9 studies), and the SAUC for placental lacunae was 0.91 (95% CI, 0.89–0.94; 13 studies). If we exclude hypervascularity and myometrial thinning, as mentioned above, the increase in the integrated SAUC was only 4%, raising it from 0.83 (0.79–0.86) to 0.87 (0.84–0.89).

Comments

This meta-analysis confirms that a quantitative evaluation of multiple sonographic markers in patients with a history of CD contributes to the prenatal evaluation of patients at risk of PAS at birth, and can be used to triage these patients and refer them to specialist centers for follow-up and multidisciplinary management.

Comparison with existing literature

This systematic review and meta-analysis of 12 cohorts and 2 case-control studies proposes a clinical-sonographic score system for prenatal evaluation of third-trimester patients at high risk of PAS. It was found that the sonographic markers of placental location in the LUS, absence of clear zone, and the presence of placental lacunae are the markers with the highest sensitivity and specificity in identifying patients with a high probability of PAS at birth. Subplacental hypervascularity and myometrial thinning slightly increased the accuracy

of these scores but placental bulging did not.

This meta-analysis confirms that a quantitative evaluation of multiple sonographic markers in patients with a history of CD contributes to the prenatal evaluation of patients at risk of PAS at birth, and can be used to triage these patients and refer them to specialist centers for follow-up and multidisciplinary management.

Strengths and limitations

Scoring systems for imaging diagnostics have been shown to support clinical decision-making and management in many medical specialties, particularly in gynecology. They include, for example, the evaluation of ovarian masses³⁵ and uterine tumors.³⁶

In a recent study, using a modified Delphi process, a consensus among experts was reached for 7 of the 11 established ultrasound markers associated with PAS, including the following: loss of clear zone; myometrial thinning; placental bulging

TABLE 3

Mean sensitivity, specificity, and summary area under the receiver operating characteristic curve for each sonographic marker and for integrated sonographic markers

Sonographic marker (# of studies)	Summary operating point			Sensitivity at fixed specificity from the ROCs	
	Mean sensitivity	Mean specificity	SAUC	90%	80%
Placental location in the LUS (13)	0.80 (0.62–0.91)	0.91 (0.71–0.98)	0.91 (0.89–0.93)	0.81	0.88
Loss of clear zone (9)	0.87 (0.75–0.94)	0.93 (0.71–0.99)	0.95 (0.71–0.99)	0.89	0.93
Myometrial thinning (7)	0.72 (0.38–0.91)	0.93 (0.71–0.99)	0.91 (0.88–0.93)	0.78	0.88
Placental lacunae (13)	0.89 (0.79–0.95)	0.78 (0.57–0.90)	0.91 (0.89–0.94)	0.80	0.88
Subplacental hypervascularity (10)	0.66 (0.42–0.84)	0.90 (0.67–0.97)	0.85 (0.81–0.88)	0.66	0.77
Integrated 6 sonographic markers	0.68 (0.53–0.80)	0.88 (0.68–0.96)	0.83 (0.79–0.86)	0.66	0.75
Integrated 4 sonographic markers	0.72 (0.52–0.86)	0.92 (0.63–0.99)	0.87 (0.84–0.89)	0.75	0.81

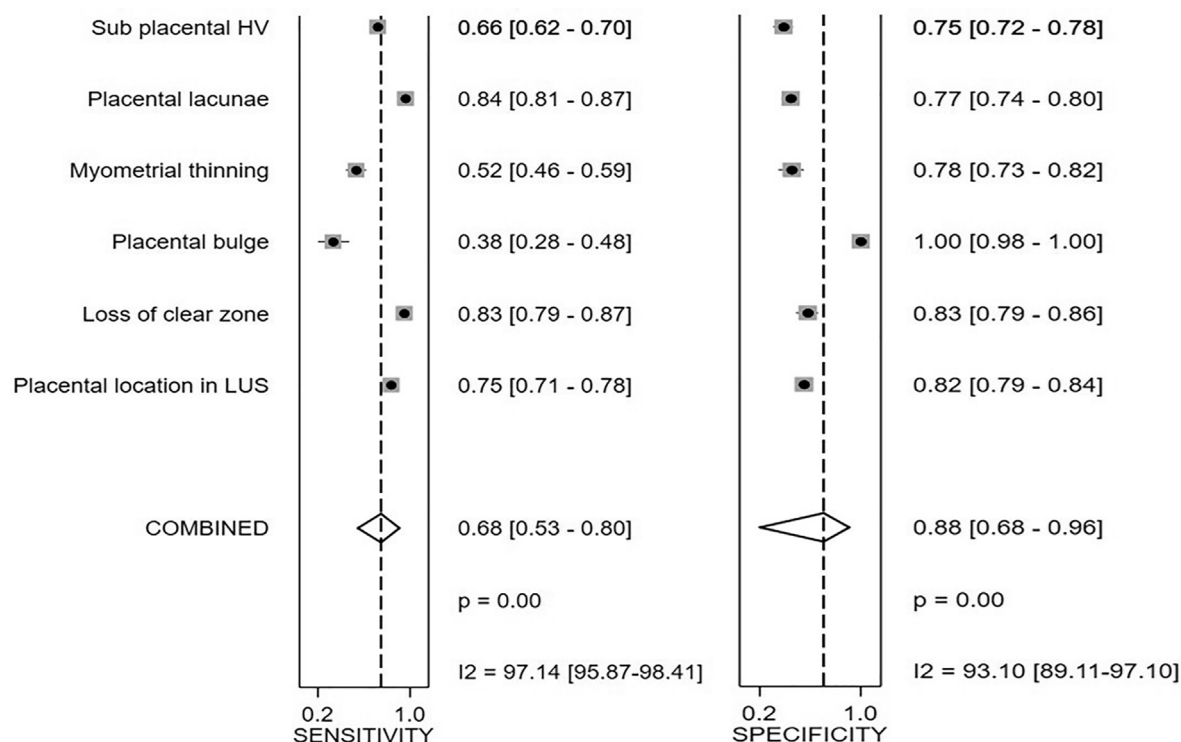
The meta-analysis of each sonographic marker and the integrated sonographic markers, including the 95% confidence intervals for the sensitivity, specificity, and SAUC, as presented in Figure 2. The columns to the right are the sensitivities at fixed specificities of 90% and 80%. The integrated value of 6 sonographic markers is obtained from the meta-analysis of the individual 5 sonographic markers as listed in the table along with placental bulging (the sixth sonographic marker). The integrated value of 4 sonographic markers is obtained from the integrated analysis of the above, excluding hypervascularity and myometrial thinning.

LUS, lower uterine segment; ROC, receiver operating characteristic curve; SAUC, summary of the area under the receiver operating characteristic curve.

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FIGURE 1

Forest plot of the 6 sonographic markers



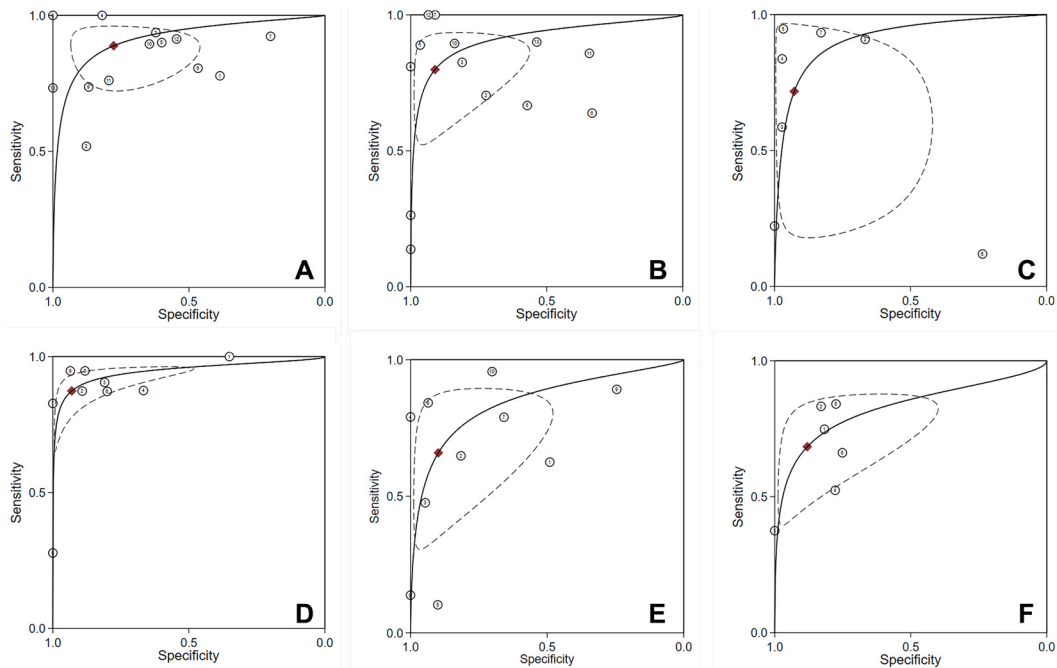
The combined sensitivities and specificities (lower line). I^2 the heterogeneity index, and P values the level of statistical significance. It is notable that the 95% confidence interval is very narrow. Subplacental HV (10 studies), placental lacunae (13 studies), myometrial thinning (7 studies), placental bulging (4 studies), loss of clear zone (9 studies), and placental location in the LUS (13 studies) are shown.

HV, hypervascularity; LUS, lower uterine segment.

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FIGURE 2

SROC of the individual sonographic markers of PAS at birth forming the clinical sonographic score system in 13 studies and their integration



The (black round circles) represent the sensitivity and specificity calculated from the individual studies. The (red rectangles) represent mean sensitivity and specificity. The (dashed circle) indicates the range of 95% confidence intervals. **A**, Placental lacunae (13 studies); **B**, placental lacunae in the lower uterine segment (13 studies); **C**, myometrial thinning (7 studies); **D**, loss of clear zone (9 studies); **E**, subplacental hypervascularity (10 studies); and **F**, the integrated SROC of all markers.

PAS, placenta accreta spectrum; SROC, summary of receiver operating characteristic curves.

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associated with bladder wall interruption; placental ballooning with exophytic mass; hypervascularity of the ureterovesical junction; placental lacunae; and bridging vessels.⁵ The first 4 markers are the consequence of uterine remodeling of the LUS after CD, and like the incidence of both placenta previa and placenta previa accreta, they increase with the number of previous CDs. The last 3 are secondary to changes in the uteroplacental and intraplacental circulations following the development of placental tissue inside a scar area.^{37–40}

All the clinical-sonographic score systems reviewed in the present study included between 3 and 6 of the established sonographic markers combined with placental location in the LUS in the

evaluation of patients with a history of CD (Table 2^{21–34}). We found that at a 10% FP rate, the sensitivity of individual ultrasound markers in predicting PAS at birth had a wide range between 66% for subplacental hypervascularity and 89% for the loss of clear zone. The corresponding sensitivities for myometrial thinning and placental lacunae were 78% and 80%, respectively (Table 3). Changes in the uterine contour following scarring of the LUS are not pathognomonic of PAS.³⁸ However, when combined with the presence of lacunae, it should alert the sonographer to the high probability of PAS at birth in patients with a history of ≥ 1 CDs presenting with a low-lying placenta or placenta previa.⁵

Clinical and research implications

The ultrasound features of the different sonographic markers predicting PAS were associated with both uterine remodeling and uteroplacental and intraplacental vascularization changes with advancing GA.^{38–41} Only 4 of the 14 studies included in the present meta-analysis reported data on the GA at ultrasound examination and at birth.^{29,28,32} In only 2 of these studies, the authors reported on the interval between the mean GA by which the ultrasound identified the prediction of PAS, and the mean GA at birth.^{22,31}

The accuracy of transabdominal scan (TAS) in the sonographic evaluation of the lower half of the uterus is limited by the distance between the probe and the

respective uterine region. Other limitations of TAS are high patient BMI and the state of bladder filling.⁵ TVS examination contributes to both prenatal management and the prediction of surgical outcomes in patients with a history of CD, regardless of whether they exhibit ultrasound markers suggestive of PAS.^{42,43} TVS is also pivotal in the differential diagnosis between a low-lying placenta and placenta previa.⁴⁴ The lack of use of a standardized classification for placental localization in the present study (Table 1^{21–34}) contributes to the high heterogeneity in diagnostic accuracy and patient outcomes among studies that included patients at high risk for PAS.

The lack of detailed data on intraoperative features and histopathologic examination used to confirm the diagnosis of PAS at birth explains the wide variation in the prevalence of PAS in general maternity population studies and its incidence in cohort studies of patients with a history of CD.^{45,46} Confirmation of the diagnosis is essential to obtain more accurate data on the quality of prenatal imaging evaluation, and on the outcomes of different management strategies in different settings. Detailed histopathologic examination may not be possible in case of conservative management, and perinatal pathologic expertise is often limited to specialist centers in high-resource countries. If the placenta can be fully detached manually during delivery (ie, without the need for partial myometrial resection) or from the fresh hysterectomy specimen, it should not be reported as PAS.⁴⁷ Intraoperative gross findings are also useful to record the severity of the condition⁴⁸ and improve the accuracy of both ultrasound markers⁴⁹ and of MRI imaging in the prenatal evaluation of patients at high risk of PAS. Only 5 studies included in the present review reported different grades of accreta placentation.^{23,24,29,32,33} In the remaining studies, the authors either reported on the gross findings at birth and/or did not describe the histologic findings (Table 1^{21–34}).

Therefore, it is essential to include in future studies the GA at which the clinical-sonographic markers were measured for calculating the sonographic score, to conduct a TVS and not a TAS assessment of the placental location, and to provide detailed data on intraoperative findings.

Conclusions

Clinical-sonographic score systems can contribute to the prenatal screening of patients at risk of PAS at birth. While we included multiple sonographic studies from the midst of pregnancy, standardized evaluation should be performed not only with strict ultrasound criteria for the placental position, mid third trimester gestational age at examination, and sonographic markers associated with PAS. Numeric sensitivities, specificities, NPVs, PPV, LR-, and LR+ should be recorded prospectively to assess their accuracy in different setups and PTP should be verified at delivery. The variables recommended for most predictive screening are: loss of clear zone underneath the placental bed, placentation in the LUS, and placenta lacunae. ■

CRedit authorship contribution statement

Marina Pekar Zlotin: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Adi Sharabi-Nov:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Hamutal Meiri:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation. **Perry Eliassi Revivo:** Writing – original draft, Methodology, Formal analysis, Data curation. **Yakkov Melcer:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Ron Maymon:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Eric Jauniaux:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

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Supplementary materials

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